

Johnston Boiler Company

Bill of Material

Assembly P/N: 11377D1

Description: Control Panel

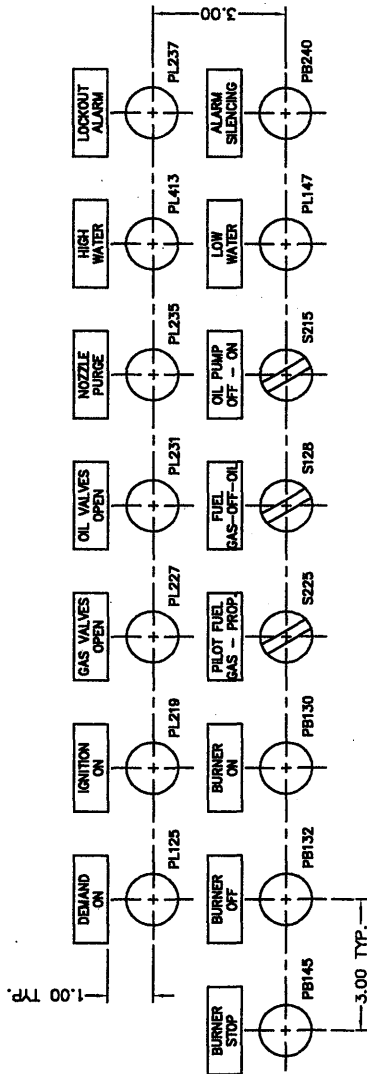
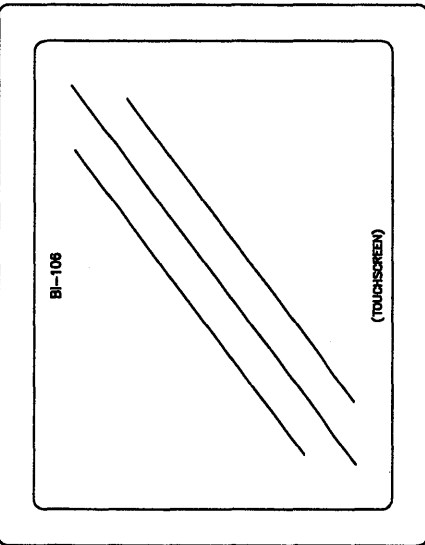
Dwg. No: 1137733201

Tag No.	Qty.	Service	Rev.	JBC	Name	Specification	Comment
-	1	FSG Communication Cable		016944	FSG CABLE COMMUNICATION 96.00	FIREYE ED512-8 W/RJ12 CONN	
BC-108	1	Expansion Hub (Combustion Controller)		037142	EXPANSION HUB CANBUS/NXC SERVO	FIREYE NXCBH - PPC6000/NX6100 EXPANSION	
XA-122	1	Alarm Horn		010029	HORN ALARM	EDWARDS 874-N5 - 120VAC	
CB126	1	PPC Unit Power Circuit Breaker		036860	CIRCUIT BREAKER 004.0A 1 POLE	EATON FAZ-C4/1-NA-SP - UL489	
CB443	1	O2 Interface Power Circuit Breaker		037363	CIRCUIT BREAKER 003.0A 1 POLE	EATON FAZ-C3/1-NA-SP - UL489	
CR126	1	PPC Demand Relay		020061	RELAY 1 POLE FORM C 10A	IDEC RH1B-UL-AC120V	
"	1	PPC Demand Relay Socket		020063	SOCKET RELAY 5 BLADE	IDEC SH1B-05C	
CR127	1	Oil Selected Relay #1		025210	RELAY 2 POLE FORM C 10A	IDEC RH2B-UL-AC110-120V	
"	1	Oil Selected Relay #1 Socket		025211	SOCKET RELAY 8 BLADE	IDEC SH2B-05C	
CR128	1	Oil Selected Relay #2		020062	RELAY 4 POLE FORM C 10A	IDEC RH4B-UL-AC120V	
"	1	Oil Selected Relay #2 Socket		020064	SOCKET RELAY 14 BLADE	IDEC SH4B-05C	
CR130	1	Burner On Relay		020061	RELAY 1 POLE FORM C 10A	IDEC RH1B-UL-AC120V	
"	1	Burner On Relay Socket		020063	SOCKET RELAY 5 BLADE	IDEC SH1B-05C	
CR134	1	Demand On Relay		020062	RELAY 4 POLE FORM C 10A	IDEC RH4B-UL-AC120V	
"	1	Demand On Relay Socket		020064	SOCKET RELAY 14 BLADE	IDEC SH4B-05C	
CR136	1	Low Fire Shutdown Relay		020062	RELAY 4 POLE FORM C 10A	IDEC RH4B-UL-AC120V	
"	1	Low Fire Shutdown Relay Socket		020064	SOCKET RELAY 14 BLADE	IDEC SH4B-05C	
CR148	1	Low Water Relay		020061	RELAY 1 POLE FORM C 10A	IDEC RH1B-UL-AC120V	
"	1	Low Water Relay Socket		020063	SOCKET RELAY 5 BLADE	IDEC SH1B-05C	
CR206	1	Blower Motor Start Relay		020061	RELAY 1 POLE FORM C 10A	IDEC RH1B-UL-AC120V	
"	1	Blower Motor Start Relay Socket		020063	SOCKET RELAY 5 BLADE	IDEC SH1B-05C	
CR217	1	Ignition Position Prove Relay		020061	RELAY 1 POLE FORM C 10A	IDEC RH1B-UL-AC120V	
"	1	Ignition Position Prove Relay Socket		020063	SOCKET RELAY 5 BLADE	IDEC SH1B-05C	
CR238	1	Lockout Alarm Relay		020061	RELAY 1 POLE FORM C 10A	IDEC RH1B-UL-AC120V	
"	1	Lockout Alarm Relay Socket		020063	SOCKET RELAY 5 BLADE	IDEC SH1B-05C	
CR240	1	Alarm Silencing Relay		020061	RELAY 1 POLE FORM C 10A	IDEC RH1B-UL-AC120V	
"	1	Alarm Silencing Relay Socket		020063	SOCKET RELAY 5 BLADE	IDEC SH1B-05C	
CR245	1	Firing Rate High Relay		020061	RELAY 1 POLE FORM C 10A	IDEC RH1B-UL-AC120V	
"	1	Firing Rate High Relay Socket		020063	SOCKET RELAY 5 BLADE	IDEC SH1B-05C	
CR247	1	Modulate Relay		020061	RELAY 1 POLE FORM C 10A	IDEC RH1B-UL-AC120V	
"	1	Modulate Relay Socket		020063	SOCKET RELAY 5 BLADE	IDEC SH1B-05C	
TDR248	1	Modulate Time Delay Relay		033914	RELAY TIME DELAY ON-DELAY	ALLEN-BRADLEY 700-FEA3TU23	
CR302	1	PPC Unit Fault Relay		020061	RELAY 1 POLE FORM C 10A	IDEC RH1B-UL-AC120V	
"	1	PPC Unit Fault Relay Socket		020063	SOCKET RELAY 5 BLADE	IDEC SH1B-05C	
CR363	1	Low Fire Hold Relay		020061	RELAY 1 POLE FORM C 10A	IDEC RH1B-UL-AC120V	
"	1	Low Fire Hold Relay Socket		020063	SOCKET RELAY 5 BLADE	IDEC SH1B-05C	
TDR367	1	Nozzle Purge Delay Off Relay		036970	RELAY TIME DELAY OFF-DELAY	ALLEN-BRADLEY 700-FEB3TU23 0.05S-10HR	
CR414	1	High Water Relay		025210	RELAY 2 POLE FORM C 10A	IDEC RH2B-UL-AC110-120V	
"	1	High Water Relay Socket		025211	SOCKET RELAY 8 BLADE	IDEC SH2B-05C	
CR542	1	Drive Running Relay (24VDC)		029561	RELAY 1 POLE FORM C 10A 24VDC	IDEC RH1B-UL-DC24V	Note: Relay has 24VDC coil.
"	1	Drive Running Relay Socket		020063	SOCKET RELAY 5 BLADE	IDEC SH1B-05C	
-	140	Terminal Blocks		035685	TERMINAL BLOCK STANDARD 8.2MM	EATON/CUTLER-HAMMER XBUT6 #24-#8 600V	
-	4	End Stops		035686	TERMINAL BLOCK END STOP	EATON/CUTLER-HAMMER XBAES35C - 35MM DIN	

Johnston Boiler Company**Bill of Material****Assembly P/N: 11377D1****Description: Control Panel****Dwg. No: 1137733201**

Tag No.	Qty.	Service	Rev.	JBC	Name	Specification	Comment
-	2	End Plates		035687	TERMINAL BLOCK END PLATE	EATON/CUTLER-HAMMER XBACUT10	
-	4	Ground Blocks		035689	TERMINAL BLOCK GROUND 8.2MM	EATON/CUTLER-HAMMER XBUT6PE #24-#8	
S121	1	Control Power Switch		035690	TERMINAL BLOCK DISCONN. 6.2MM	EATON/CUTLER-HAMMER XBUT4MT #26-#10 600V	
-	14	Marking Tags		035691	TAGS TERM. BLOCK BLANK 8.2MM	EATON/CUTLER-HAMMER XBMZB8 - QTY 10/UNIT	
-	1	Enclosure		033390	ENCLOSURE NEMA4/12/13 42X36X10	C-SD423610 HOFFMAN	
-	1	Panel		021283	PANEL SUB 42X36	C-P4236 HOFFMAN CONCEPT	
Notes							
1) Ref. P/N 11378D1							
Rev Data	No.	Date	By	Chk.	Description	Data Entry	
	0	20-Sep-17	TJN	TJA	-		

BI-102



42.00

(continued)

36.00



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CAD/FULL
DATE
9-20-17
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T. NINK
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APPROVED
TJN

JOHNSTON BOILER COMPANY
FERRYSBURG, MICHIGAN 49409

CONTROL PANEL LAYOUT
FIREYE YB110/YZ300/PPC6000 - HG

DWG. NO. 1137733201

(PAGE 1 OF 2)

SO#

NOTES:



Abstract

JOHNSTON BOILER COMPANY
FERRYSBURG, MICHIGAN 49409

D	DWG. NO.	1137733201
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(PAGE 2 OF 2) REV

Johnston Boiler Company

Bill of Material

Assembly P/N: 11377D2

Description: Power Panel - 460V
100(VFD)/25/3HP 24kW

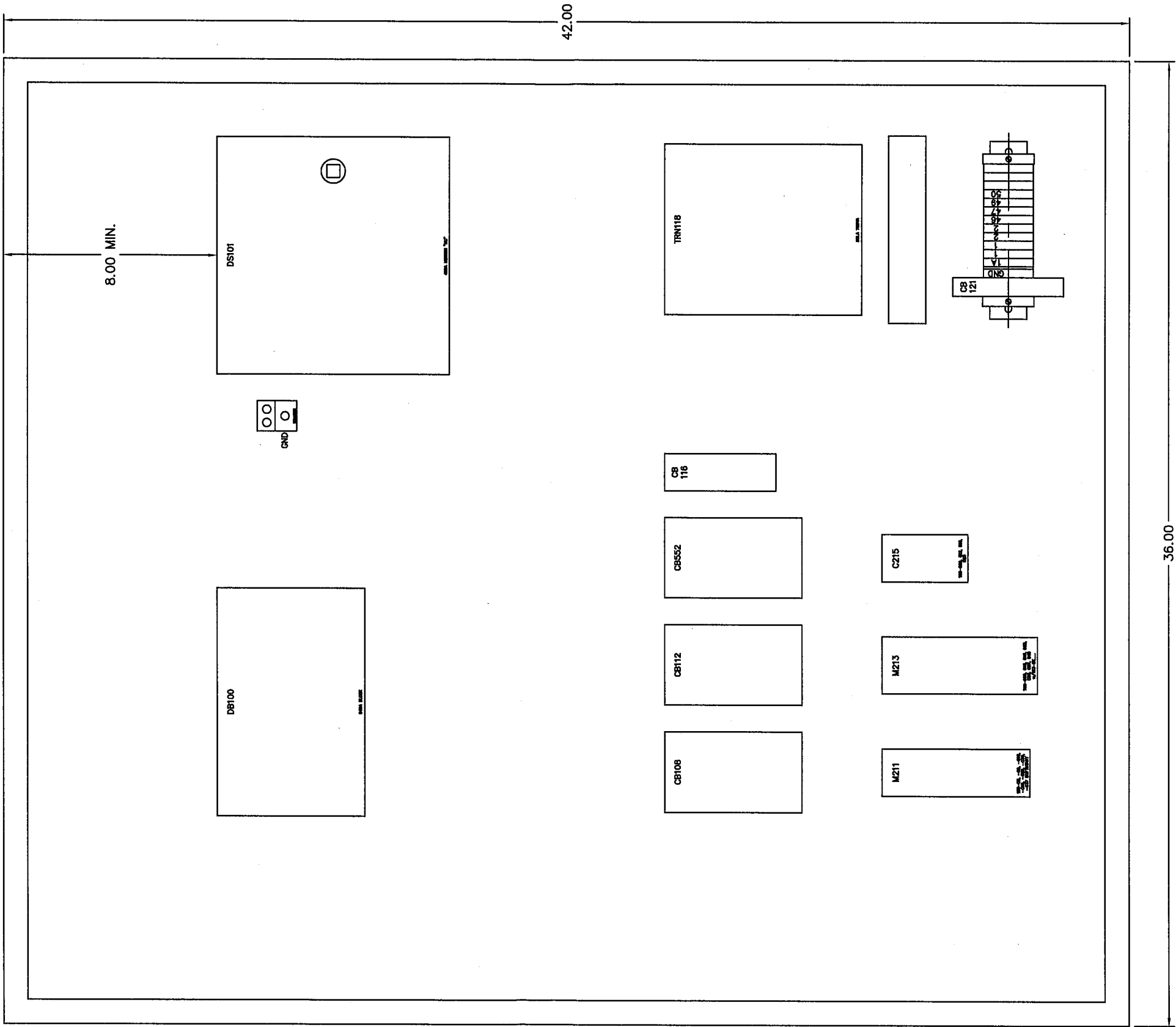
Dwg. No: 1137735701

Tag No.	Qty.	Service	Rev.	JBC	Name	Specification	Comment
DS101	1	Main Disconnect Switch Body		039398	SWITCH DISCONNECT 400A	MERSEN M400U30 - 600V NON-FUSIBLE UL98	
"	1	Main Disconnect Switch Operator		039399	OPERATOR DISC SWITCH NEMA4/4X	MERSEN HB125X - BLACK	
"	1	Main Disconnect Switch Shaft		039400	SHAFT DISC SWITCH	MERSEN SFB325 - 12.7IN	
"	1	Main Disconnect Terminal Lugs		039414	LUG KIT DISCONNECT #2-600MCM	MERSEN LUG400 (SET OF 6)	
-	1	Ground Lug		033899	LUG TERMINAL #14-1/0	BURNDY K2A25U 2-WIRE	
DB100	1	Distribution Block		016437	BLOCK DISTRIBUTION 840A 3-POLE	BUSSMANN 16528-3 (2)600MCM-#2X(4)#4-14 &	
CB108	1	Compressor Motor Circuit Breaker		040162	CIRCUIT BREAKER 090A 3 POLE	ALLEN-BRADLEY 140G-G2C3-C90 - UL489	
CB112	1	Oil Pump Motor Circuit Breaker		040163	CIRCUIT BREAKER 040A 3 POLE	ALLEN-BRADLEY 140G-G2C3-C40 - UL489	
CB552	1	Oil Heater Circuit Breaker		040164	CIRCUIT BREAKER 015A 3 POLE	ALLEN-BRADLEY 140G-G2C3-C15 - UL489	
-	6	Circuit Breaker Lug Kits		039899	LUG KIT CIRCUIT BREAKER	ALLEN-BRADLEY 140G-G-TLC13 - #14-1/0 FOR	
CB116	1	Transformer Primary Circuit Breaker		040165	CIRCUIT BREAKER 008.0A 2 POLE	EATON FAZ-C8/2-NA - UL489	
CB121	1	Transformer Secondary Circuit Breaker		036688	CIRCUIT BREAKER 008.0A 1 POLE	EATON FAZ-C8/1-NA-SP - UL489	
TRN118	1	Control Transformer		037179	TRANSFORMER 230/460V X 115V	SOLA/HEVI-DUTY E750 750VA NO FUSE BLOCKS	
IZU-800	1	Variable Frequency Drive		035974	INVERTER PKG NEMA12 W/DISCONN	DANFOSS/FIREYE NXVFD1004 100HP N	See notes 1 & 2 below.
M211	1	Compressor Motor Soft Starter		040166	CONTROLLER MOTOR 037A	ALLEN-BRADLEY 150-C37NBD SOFT START	
M213	1	Oil Pump Motor Contactor		027640	CONTACTOR IEC 9A	ALLEN-BRADLEY 100-C09D10	
"	1	Oil Pump Motor Overload Relay		035398	RELAY OL IEC 3.2-16A	ALLEN-BRADLEY 193-EEDB (E1 PLUS)	
C215	1	Oil Heater Contactor		027642	CONTACTOR IEC 16A	ALLEN-BRADLEY 100-C16D10	
-	12	Terminal Blocks		035685	TERMINAL BLOCK STANDARD 8.2MM	EATON/CUTLER-HAMMER XBUT6 #24-#8 600V	
-	20'	Wire		040167	WIRE THHN 250MCM BLACK	STRANDED, UL LISTED	
-	2	End Stops		035686	TERMINAL BLOCK END STOP	EATON/CUTLER-HAMMER XBAES35C - 35MM DIN	
-	1	End Plate		035687	TERMINAL BLOCK END PLATE	EATON/CUTLER-HAMMER XBACUT10	
-	1	Ground Block		035689	TERMINAL BLOCK GROUND 8.2MM	EATON/CUTLER-HAMMER XBUT6PE #24-#8	
-	2	Marking Tags		035691	TAGS TERM. BLOCK BLANK 8.2MM	EATON/CUTLER-HAMMER XBMZB8 - QTY 10/UNIT	
-	1	Enclosure		033390	ENCLOSURE NEMA4/12/13 42X36X10	C-SD423610 HOFFMAN	
-	1	Panel		021283	PANEL SUB 42X36	C-P4236 HOFFMAN CONCEPT	
-	1	Data Pocket		026479	DATA POCKET LARGE (12X12)	ADP2 HOFFMAN	
-	1	Nameplate		016573	NAMEPLATE PANEL ELECTRICAL	JOHNSTON BOILER COMPANY 8.0X6.0 METAL	

Notes

- 1) Item is mounted in separate enclosure
- 2) Item ships loose after factory firetest.

Rev Data	No.	Date	By	Chk.	Description	Data Entry
	0	19-Sep-17	TJN	TJN	-	

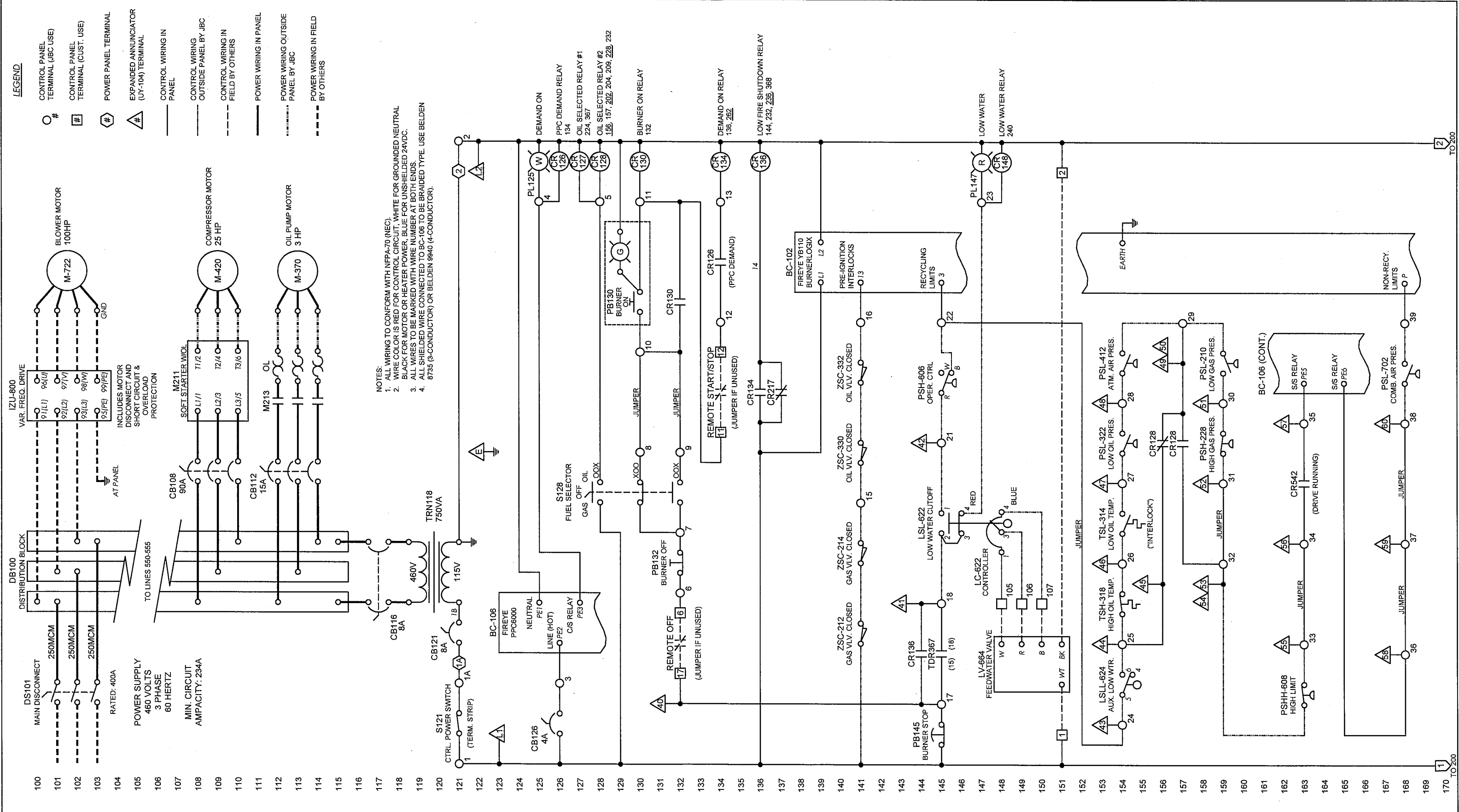


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DATE 9-19-17
DRAWN T. NINK
CHECKED
APPROVED TJN

JOHNSTON BOILER COMPANY		ISO#
FERRYSBURG, MICHIGAN 49409		
POWER PANEL LAYOUT		
100HP(VFD)/37A (SOFT)/9-43A/9-23A		
DWG. NO.	1137735701	REV

REV	DESCRIPTION



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SCALE NONE	JOHNSTON BOILER COMPANY		SO#
DATE 9-19-17	FERRYSBURG, MICHIGAN 49409		
DRAWN T. NINK	WIRING SCHEMATIC		
CHECKED	YB110/YZ300/PPC6000 - HG		
APPROVED TJN	DWG. NO. 1137730801	(PAGE 1 OF 5)	REV

